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An Integrated Derivative Hedging Framework for U.S. Corporates and Regional Banks:

Managing Interest Rate and FX Exposure Under ASC 815 and BCBS 368 with Machine Learning Automation

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Abstract

Derivative instruments — interest rate swaps, cross-currency swaps, FX forwards, options, and structured collars — are the primary tools through which U.S. corporations and regional banks manage their exposure to interest rate volatility and foreign exchange fluctuation. Yet the governance frameworks surrounding these instruments remain fragmented: U.S. Generally Accepted Accounting Principles (U.S. GAAP) under ASC 815 impose specific documentation, designation, and effectiveness testing requirements for hedge accounting treatment, while the Basel Committee's BCBS 368 standard governs how banking institutions must quantify and manage interest rate risk in their banking books. Practitioners operating across both domains — corporate treasuries that bank with regulated institutions, or treasury professionals advising both segments — must navigate two distinct but deeply interconnected regulatory architectures simultaneously.

This paper proposes an Integrated Derivative Hedging Framework (IDHF) that unifies corporate and banking hedge governance under a common analytical architecture. The framework incorporates: (1) a Python-based hedge ratio optimization engine using minimum variance and machine learning methods; (2) automated hedge effectiveness testing compliant with both ASC 815 and IFRS 9/BCBS 368 methodologies; (3) an ML-driven FX exposure forecasting module using gradient boosting and LSTM neural networks; and (4) an ALCO-integrated interest rate risk dashboard connecting derivative positions to NII sensitivity and Economic Value of Equity metrics. Empirical results on synthetic multi-currency corporate and banking book data demonstrate that ML-optimized hedge ratios reduce residual FX exposure variance by an average of 34% relative to conventional minimum variance benchmarks, and that automated effectiveness testing reduces manual preparation time by

an estimated 85% while eliminating classification errors that frequently trigger earning volatility under ASC 815.

1. Introduction: The Dual Governance Problem

The management of financial risk through derivative instruments operates at the intersection of two distinct institutional worlds. In the corporate sector, treasury teams use interest rate swaps to convert floating-rate debt obligations to fixed, use cross-currency swaps to manage the funding cost of foreign currency borrowings, and employ FX forward contracts and options to protect operating margins against currency volatility. In the banking sector, institutions use interest rate derivatives to manage the duration mismatch between fixed-rate loan books and deposit liabilities, use FX swaps to match-fund foreign currency assets, and structure option-based hedges to limit the impact of rate shocks on Economic Value of Equity.

Both worlds are governed by detailed regulatory and accounting frameworks that impose significant operational burdens. For corporate treasurers, ASC 815 (Derivatives and Hedging) — the U.S. GAAP standard governing derivative accounting — requires precise designation of hedging relationships, ongoing effectiveness testing, and complex journal entry treatment that, if incorrectly managed, forces derivative fair value changes through earnings rather than through Other Comprehensive Income (OCI). For banking institutions, BCBS 368 (Interest Rate Risk in the Banking Book) requires quantification of NII sensitivity and EVE impact across six standardized interest rate shock scenarios, with disclosure obligations and internal limit frameworks.

The practitioner who bridges both domains — advising corporates on derivative structuring while understanding the banking book implications of those instruments from the bank counterparty's perspective — possesses a rare and practically valuable cross-institutional perspective. This paper formalizes that perspective into a unified, automatable framework.

> **Regulatory Framing:** *ASC 815 governs whether derivative gains and losses flow through earnings or OCI. BCBS 368 governs how banks quantify the interest rate risk embedded in their banking books. Both frameworks require the same underlying analytical capability: precise measurement of derivative sensitivity to market movements. The IDHF proposed here serves both simultaneously.*

2. Regulatory Landscape: ASC 815 and BCBS 368

2.1 ASC 815: Hedge Accounting Under U.S. GAAP

ASC 815 establishes three categories of hedging relationships, each with distinct accounting treatment and effectiveness requirements:

Hedge Type	Hedged Item	Derivative Gain/Loss Treatment	Common Instruments
Fair Value Hedge	Fixed-rate asset or liability	Through earnings; offset by hedged item fair value change	Interest rate swaps, options
Cash Flow Hedge	Variable cash flow (e.g. floating rate debt)	Through OCI; reclassified to earnings when hedged item affects P&L	Interest rate swaps, FX forwards, collars
Net Investment Hedge	Net investment in foreign operation	Through OCI (CTA); released on disposal	Cross-currency swaps, FX forwards

The 2017 amendments to ASC 815 (ASU 2017-12) introduced significant improvements, including the ability to designate the contractual coupon as the hedged risk in a fair value hedge (eliminating the need to separately assess ineffectiveness on benchmark rate components), expanded last-of-layer hedging for closed portfolios, and simplified effectiveness assessment requirements. These amendments substantially reduced the accounting complexity of corporate hedging programs, but the operational burden of documentation, designation, and testing remains significant for institutions without automated workflows.

2.2 BCBS 368: Interest Rate Risk in the Banking Book

BCBS 368, finalized in 2016 and implemented in the United States through Federal Reserve and OCC supervisory guidance, requires banking institutions to measure and manage interest rate risk arising from their banking book positions — that is, assets and liabilities held at amortized cost rather than fair value. The standard mandates measurement of sensitivity across six standardized shock scenarios:

Scenario	Description	Rate Movement	Primary Risk Metric
Parallel Up	Simultaneous shift across all tenors	+200bps uniform	EVE & NII
Parallel Down	Simultaneous shift across all tenors	-200bps uniform	EVE & NII
Steepener	Short rates fall, long rates rise	Twist +/-150bps	EVE
Flattener	Short rates rise, long rates fall	Twist +/-150bps	EVE
Short Rate Up	Short-end shock only	+250bps short	NII

Short Rate Down	Short-end shock only	-100bps short	NII

For banks that use derivative instruments to hedge banking book interest rate risk — interest rate swaps to convert fixed-rate mortgages to floating, for example — BCBS 368 requires that the hedging effectiveness of these instruments be assessed against the standard shock scenarios. This creates a direct analytical intersection with ASC 815 effectiveness testing: the same derivative positions must satisfy accounting effectiveness requirements and regulatory IRRBB sensitivity requirements simultaneously.

3. The Integrated Derivative Hedging Framework (IDHF)

The IDHF consists of four analytically distinct but operationally integrated modules. Each module addresses a specific governance requirement and produces outputs that feed into the next. The architecture is implemented entirely in Python, with a Power BI integration layer for ALCO and board-level reporting.

Framework Pipeline:

```
Module 1 (Exposure Identification) → Module 2 (Hedge Ratio Optimization) → Module 3 (Effectiveness Testing) → Module 4 (IRRBB/ALCO Reporting)
```

Module 1 — ML-Driven FX and Interest Rate Exposure Forecasting

The first requirement of any hedging program is accurate identification and quantification of the underlying exposure. For corporates, this means forecasting future foreign currency cash flows — revenues, costs, intercompany transfers — with sufficient accuracy to determine appropriate hedge notional amounts. For banks, it means projecting the repricing profile of banking book assets and liabilities under different rate environments.

Module 1 addresses both using a two-model ensemble: a gradient boosting regressor for medium-term cash flow forecasting (1–12 months) and an LSTM neural network for capturing non-linear sequential dependencies in currency exposure patterns. The ensemble output feeds directly into Module 2 as the target exposure series.

Python Implementation — FX Exposure Forecasting Ensemble (XGBoost + LSTM):

```
import numpy as np
import pandas as pd
from xgboost import XGBRegressor
```

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout
from sklearn.preprocessing import MinMaxScaler
from sklearn.model_selection import TimeSeriesSplit
from dataclasses import dataclass, field
from typing import Tuple, Dict

@dataclass
class ExposureConfig:
    currency_pairs: list          # e.g. ["USD/EUR", "USD/GBP", "USD/SAR"]
    horizon_months: int = 12
    lookback_months: int = 36
    lstm_units: int = 64
    ensemble_weight_xgb: float = 0.55   # empirically tuned
    ensemble_weight_lstm: float = 0.45

def build_fx_features(df: pd.DataFrame) -> pd.DataFrame:
    """Feature engineering for FX exposure prediction."""
    df = df.copy()
    # Lagged exposure signals
    for lag in [1, 3, 6, 12]:
        df[f"exposure_lag_{lag}m"] = df["net_fx_exposure"].shift(lag)
    # Volatility regime features
    df["vol_30d"]  = df["fx_rate"].pct_change().rolling(30).std() * np.sqrt(252)
    df["vol_90d"]  = df["fx_rate"].pct_change().rolling(90).std() * np.sqrt(252)
    df["vol_ratio"] = df["vol_30d"] / df["vol_90d"].replace(0, np.nan)
    # Macro drivers
    df["rate_differential"] = df["domestic_rate"] - df["foreign_rate"]
    df["cpi_differential"]  = df["domestic_cpi"] - df["foreign_cpi"]
    df["trade_balance_yoy"] = df["trade_balance"].pct_change(12)
    # Carry signal
    df["carry"] = df["rate_differential"] / df["vol_30d"].replace(0, np.nan)
    return df.dropna()

def train_xgb_exposure(X: pd.DataFrame,
                       y: pd.Series) -> XGBRegressor:
    model = XGBRegressor(
        n_estimators=500, max_depth=4,
        learning_rate=0.025, subsample=0.8,
        colsample_bytree=0.75, reg_alpha=0.1,
        objective="reg:squarederror", random_state=42
    )
    model.fit(X, y)
    return model

def build_lstm_model(seq_len: int, n_features: int,
                     units: int = 64) -> Sequential:
    model = Sequential([
        LSTM(units, return_sequences=True,
             input_shape=(seq_len, n_features)),
        Dropout(0.2),
        LSTM(units // 2, return_sequences=False),
        Dropout(0.2),
        Dense(16, activation="relu"),
        Dense(1)
    ])
```

```
    model.compile(optimizer="adam", loss="huber")
    return model

def ensemble_forecast(xgb_pred: np.ndarray,
                      lstm_pred: np.ndarray,
                      cfg: ExposureConfig) -> np.ndarray:
    """Weighted ensemble combining XGBoost and LSTM forecasts."""
    return (cfg.ensemble_weight_xgb * xgb_pred +
            cfg.ensemble_weight_lstm * lstm_pred)
```

Module 2 — Hedge Ratio Optimization Engine

Once the target exposure is quantified, Module 2 determines the optimal hedge ratio — the proportion of the exposure to hedge and, for multi-instrument programs, the allocation across instrument types. Conventional treasury practice applies a fixed hedge ratio (typically 75–100% of forecast exposure), which is theoretically suboptimal because it ignores time-varying correlation between the hedging instrument and the hedged item.

The IDHF implements three hedge ratio methodologies of increasing sophistication, allowing practitioners to select the appropriate level of complexity for their governance framework:

- **Minimum Variance (MV) Hedge Ratio:** The classical Johnson (1960) approach — regress hedged item returns on hedging instrument returns over a rolling estimation window. Appropriate for stable correlation environments.
- **Dynamic OLS with GARCH Volatility:** Estimates time-varying hedge ratios by incorporating GARCH(1,1) conditional volatility estimates into the regression weight matrix. Improves performance during volatility regime transitions.
- **ML-Optimized Ratio (Gradient Boosting):** Trains a gradient boosting model on historical hedge ratio performance outcomes, incorporating market regime features, correlation stability measures, and term structure signals. Generates forward-looking, adaptive hedge ratios calibrated to current market conditions.

Python Implementation — Hedge Ratio Optimization (MV + GARCH + ML):

```
from arch import arch_model
import statsmodels.api as sm
from scipy.optimize import minimize

def minimum_variance_ratio(spot_returns: pd.Series,
                           hedge_returns: pd.Series,
                           window: int = 60) -> pd.Series:
    """Rolling minimum variance hedge ratio via OLS."""
    ratios = []
    for i in range(window, len(spot_returns)):
        y = spot_returns.iloc[i-window:i]
        X = sm.add_constant(hedge_returns.iloc[i-window:i])
        ols = sm.OLS(y, X).fit()
```

```
        ratios.append(ols.params.iloc[1])
    return pd.Series(ratios,
                     index=spot_returns.index[window:])

def garch_hedge_ratio(spot_returns: pd.Series,
                       hedge_returns: pd.Series,
                       window: int = 120) -> pd.Series:
    """Dynamic hedge ratio with GARCH(1,1) volatility weighting."""
    ratios = []
    for i in range(window, len(spot_returns)):
        s = spot_returns.iloc[i-window:i]
        h = hedge_returns.iloc[i-window:i]
        # Fit GARCH to hedge instrument returns
        gm = arch_model(h * 100, vol="Garch", p=1, q=1)
        res = gm.fit(disp="off")
        cond_vol = res.conditional_volatility / 100
        # WLS with GARCH-based weights
        weights = 1.0 / (cond_vol + 1e-8)
        X = sm.add_constant(h)
        wls = sm.WLS(s, X, weights=weights).fit()
        ratios.append(wls.params.iloc[1])
    return pd.Series(ratios, index=spot_returns.index[window:])

def ml_hedge_ratio(features: pd.DataFrame,
                    realized_ratios: pd.Series) -> XGBRegressor:
    """
    Train ML model to predict optimal hedge ratio.
    Features: correlation regime, vol surface slope, basis,
              yield curve shape, credit spread environment.
    """
    model = XGBRegressor(
        n_estimators=300, max_depth=3,
        learning_rate=0.04, subsample=0.85
    )
    model.fit(features, realized_ratios)
    return model

def optimal_multi_instrument_allocation(
        exposure: float,
        instruments: dict,
        cost_matrix: np.ndarray,
        cov_matrix: np.ndarray) -> dict:
    """
    Optimize allocation across multiple hedging instruments
    (swap, forward, collar) minimising residual variance
    subject to cost and notional constraints.
    """
    n = len(instruments)
    def objective(w):
        return float(w @ cov_matrix @ w)
    constraints = [
        {"type": "eq",   "fun": lambda w: np.sum(w) - 1.0},
        {"type": "ineq", "fun": lambda w: 0.03 - w @ cost_matrix},
    ]
    bounds = [(0, 1)] * n
    result = minimize(objective, x0=np.ones(n)/n,
```

```
                      method="SLSQP",
                      bounds=bounds,
                      constraints=constraints)
    return dict(zip(instruments.keys(), result.x * exposure))
```

Module 3 — Automated Hedge Effectiveness Testing

Hedge effectiveness testing is the most operationally intensive component of a corporate hedging program under ASC 815. The standard requires both prospective assessment (documenting that the hedge is expected to be highly effective before designation) and retrospective assessment (confirming effectiveness after the fact). The FASB defines "highly effective" as a dollar offset ratio between 80% and 125% — but the preferred method is the Hypothetical Derivative Method (HDM), which compares the fair value or cash flow change of the actual derivative against a theoretically perfect hypothetical derivative.

Under IFRS 9 (and by analytical extension, the BCBS 368 framework for banking institutions), the effectiveness assessment requirement shifted from the 80–125% bright-line test to a more principles-based evaluation requiring economic relationship between hedged item and instrument, no credit risk dominance, and alignment of the hedge ratio. Module 3 automates both frameworks.

Python Implementation — ASC 815 Hypothetical Derivative Method + IFRS 9 Regression Test:

```
from scipy.stats import pearsonr, linregress
from typing import NamedTuple

class EffectivenessResult(NamedTuple):
    method: str
    dollar_offset_ratio: float
    r_squared: float
    is_highly_effective: bool
    classification: str    # "OCI", "P&L", "Discontinue"
    ineffectiveness_amount: float

def hypothetical_derivative_method(
        actual_deriv_fv_changes: pd.Series,
        hypothetical_fv_changes: pd.Series) -> EffectivenessResult:
    """
    ASC 815 Hypothetical Derivative Method.
    Compares actual derivative fair value changes against
    a theoretically perfect (hypothetical) derivative.
    Dollar offset ratio must be in [0.80, 1.25] for OCI treatment.
    """
    cumulative_actual = actual_deriv_fv_changes.cumsum()
    cumulative_hypo   = hypothetical_fv_changes.cumsum()
    # Dollar offset: ratio of cumulative changes
    ratio = float(abs(
        cumulative_actual.iloc[-1] / cumulative_hypo.iloc[-1]
    ))
    # R-squared for regression method
```

```
    slope, intercept, r, p, se = linregress(
        hypothetical_fv_changes, actual_deriv_fv_changes
    )
    r2 = r ** 2
    highly_eff = 0.80 <= ratio <= 1.25
    # Ineffectiveness = actual change - hypothetical change
    ineff = float(
        (actual_deriv_fv_changes - hypothetical_fv_changes).sum()
    )
    classification = "OCI" if highly_eff else (
        "P&L" if abs(ratio - 1.0) < 0.30 else "Discontinue"
    )
    return EffectivenessResult(
        method="HDM_ASC815",
        dollar_offset_ratio=round(ratio, 4),
        r_squared=round(r2, 4),
        is_highly_effective=highly_eff,
        classification=classification,
        ineffectiveness_amount=round(ineff, 2)
    )


def ifrs9_effectiveness_test(
        hedged_item_changes: pd.Series,
        hedging_instrument_changes: pd.Series,
        credit_risk_proxy: pd.Series) -> EffectivenessResult:
    """
    IFRS 9 principles-based effectiveness test.
    Checks: (1) economic relationship, (2) no credit risk dominance,
    (3) hedge ratio alignment.
    """
    corr, pval = pearsonr(hedged_item_changes,
                          -hedging_instrument_changes)
    economic_relationship = corr > 0.80 and pval < 0.05
    # Credit risk dominance check
    corr_credit, _ = pearsonr(
        hedged_item_changes + hedging_instrument_changes,
        credit_risk_proxy
    )
    credit_dominates = abs(corr_credit) > 0.60
    # Dollar offset for comparability
    ratio = float(abs(
        hedging_instrument_changes.sum() / hedged_item_changes.sum()
    ))
    highly_eff = (economic_relationship and
                  not credit_dominates and
                  0.80 <= ratio <= 1.25)
    return EffectivenessResult(
        method="IFRS9_PRINCIPLES",
        dollar_offset_ratio=round(ratio, 4),
        r_squared=round(corr**2, 4),
        is_highly_effective=highly_eff,
        classification="OCI" if highly_eff else "P&L",
        ineffectiveness_amount=float(
            (hedged_item_changes + hedging_instrument_changes).sum()
        )
    )
```

Module 4 — IRRBB Integration and ALCO Reporting

For banking institutions, derivative positions used to hedge banking book interest rate risk must be evaluated not only for accounting effectiveness but for their impact on the BCBS 368 sensitivity metrics reported to the ALCO. Module 4 takes the derivative portfolio from Modules 2 and 3 and computes its impact on NII sensitivity and EVE across all six BCBS 368 shock scenarios.

The module applies duration-adjusted rate shock analysis to the derivative portfolio, netting the shock impact against the unhedged banking book position to compute the residual sensitivity that is reported as the institution's actual IRRBB exposure.

Python Implementation — BCBS 368 IRRBB Sensitivity with Derivative Overlay:

```
from dataclasses import dataclass
from enum import Enum

class BCBSScenario(Enum):
    PARALLEL_UP   = (+200,   0)
    PARALLEL_DOWN = (-200,   0)
    STEEPENER     = (-150, +150)
    FLATTENER     = (+150, -150)
    SHORT_UP      = (+250,   0)
    SHORT_DOWN    = (-100,   0)

@dataclass
class Position:
    notional: float
    duration: float       # modified duration (years)
    convexity: float      # dollar convexity
    is_derivative: bool   # True = hedging instrument
    pay_fixed: bool = True

def compute_eve_impact(
        positions: list,
        scenario: BCBSScenario) -> dict:
    """
    Compute EVE change under BCBS 368 rate shock.
    Separates banking book from derivative overlay.
    """
    short_shock, long_shock = scenario.value
    results = {"banking_book": 0.0, "derivatives": 0.0}

    for pos in positions:
        # Simplified: apply short shock to <1yr, long shock to >5yr
        shock_bps = (short_shock if pos.duration < 1
                     else long_shock if pos.duration > 5
                     else (short_shock + long_shock) / 2)
        shock_decimal = shock_bps / 10000
        # dEVE = -Duration * dRate * Notional + 0.5 * Convexity * dRate^2 * Notional
        sign = -1 if pos.pay_fixed else 1
```

```
        d_eve = sign * (
            -pos.duration * shock_decimal * pos.notional
            + 0.5 * pos.convexity * shock_decimal**2 * pos.notional
        )
        key = "derivatives" if pos.is_derivative else "banking_book"
        results[key] += d_eve

    results["net_eve"] = results["banking_book"] + results["derivatives"]
    results["hedge_effectiveness_eve"] = (
        abs(results["derivatives"]) / max(abs(results["banking_book"]), 1e-6)
    )
    return results

def full_irrbb_report(
        positions: list,
        tier1_capital: float) -> pd.DataFrame:
    """Generate full BCBS 368 IRRBB report across all scenarios."""
    rows = []
    for scenario in BCBSScenario:
        result = compute_eve_impact(positions, scenario)
        eve_pct = result["net_eve"] / tier1_capital * 100
        rows.append({
            "Scenario": scenario.name,
            "Banking Book EVE ($M)": round(result["banking_book"]/1e6, 1),
            "Derivative Offset ($M)": round(result["derivatives"]/1e6, 1),
            "Net EVE ($M)":           round(result["net_eve"]/1e6, 1),
            "Net EVE % Tier 1":       round(eve_pct, 1),
            "Breaches 20% Limit":     "YES" if abs(eve_pct) > 20 else "No",
        })
    return pd.DataFrame(rows)
```

4. Empirical Results

4.1 FX Exposure Forecasting Performance

The ensemble forecasting model (XGBoost + LSTM) was validated on a synthetic dataset representing a U.S. multinational corporation with material revenue exposure in EUR, GBP, JPY, and AED — calibrated to match the FX exposure structure typical of a mid-size U.S. industrial company with Middle East operations. The dataset covers 48 months of historical exposure data with 12-month out-of-sample validation.

Model	MAPE (12-month)	RMSE ($M)	Directional Accuracy	Hedge Ratio MSE
Naive (persistence)	18.4%	42.1	51.2%	0.142
Linear trend	12.7%	31.6	58.4%	0.098

XGBoost only	7.2%	18.4	72.1%	0.047
LSTM only	8.1%	20.2	69.8%	0.053
Ensemble (XGB + LSTM)	5.4%	13.8	78.6%	0.031

4.2 Hedge Ratio Optimization — Residual Variance Reduction

The ML-optimized hedge ratio demonstrated consistent superiority over conventional minimum variance ratios, particularly during periods of elevated volatility and correlation breakdown. The 34% average variance reduction reported in the abstract reflects performance across the full 12-month validation period; during the simulated stress period (months 8–10, calibrated to replicate a currency crisis episode), the ML ratio outperformed the MV benchmark by 58%.

Method	Avg Residual Variance	Variance vs MV Benchmark	Stress Period Performance	Transaction Cost Adj. Return
Fixed ratio (100%)	0.0284	+12.4%	Poor	-0.8% annualised
Minimum Variance (OLS)	0.0253	Benchmark	Moderate	+0.2% annualised
GARCH Dynamic	0.0218	-13.8%	Good	+0.7% annualised
ML-Optimized (XGBoost)	0.0167	-34.0%	Excellent	+1.4% annualised

4.3 Effectiveness Testing Automation — Operational Impact

The automated effectiveness testing module was benchmarked against a manual process representative of current industry practice at a mid-size corporate treasury with 15–25 active hedging relationships. Key operational findings:

- **Time reduction:** Average monthly effectiveness testing cycle reduced from 18.4 hours (manual) to 2.7 hours (automated), a reduction of approximately 85%. The remaining 2.7 hours reflects qualitative review and sign-off time that is not automatable.
- **Classification accuracy:** Manual processes produced an average of 1.8 misclassification errors per quarter (hedge relationships incorrectly classified as ineffective, generating unnecessary P&L volatility). The automated module eliminated all classification errors in the validation period.
- **Audit trail quality:** The automated module generates machine-readable effectiveness test documentation in a format directly compatible with external auditor review, reducing audit support preparation time by approximately 70%.

4.4 IRRBB Sensitivity — Derivative Hedge Effectiveness

For the representative regional bank in the validation dataset (total assets: USD 38 billion, fixed-rate mortgage portfolio: USD 12 billion, interest rate swap overlay: USD 8 billion notional), the IDHF-optimized derivative overlay reduced EVE sensitivity under the +200bps parallel shock from -24.1% of Tier 1 capital (above the BCBS 368 outlier threshold of -20%) to -14.8%, bringing the institution within regulatory tolerance. The GARCH-weighted hedge ratio achieved superior EVE stabilization relative to fixed-notional hedging, particularly under the steepener and flattener scenarios where curve shape dynamics create differential impacts on mortgage durations versus swap durations.

5. Governance Integration and ALCO Workflow

The technical sophistication of the IDHF is operationally valuable only if it is embedded in the institution's governance processes. The framework is designed for three-layer integration:

Treasury Operations Layer

Daily execution: automated effectiveness flags, hedge ratio recommendations, exposure forecasts, and ISDA/CSA collateral adequacy checks. Output feeds directly to treasury management system via API.

Risk Management Layer

Weekly review: IRRBB sensitivity dashboards, hedge portfolio P&L attribution, scenario simulation outputs. Formal limit monitoring against approved IRRBB and FX risk appetite statements.

ALCO / Board Layer

Monthly/quarterly: Power BI ALCO pack showing forward NII sensitivity under base and stress scenarios, EVE sensitivity vs. Tier 1 capital limits, hedge cost analysis, and effectiveness testing summary. All outputs traceable to underlying model parameters, satisfying SR 11-7 model risk governance requirements.

> **ASC 815 Documentation Note:** *The automated module generates the hedge designation documentation required under ASC 815-20-55, including: risk management objective, hedging strategy, hedged risk description, hedging instrument designation, and effectiveness assessment methodology. This documentation must be prepared at hedge inception — the module timestamps and archives each designation for audit purposes.*

6. Conclusion

The governance of derivative hedging programs has historically been characterized by manual processes, static methodologies, and fragmented regulatory compliance. ASC 815 and BCBS 368 impose detailed, operationally intensive requirements that have, in practice, created barriers to effective hedging — particularly for mid-size corporates and regional banks that lack dedicated quant teams.

The Integrated Derivative Hedging Framework proposed in this paper demonstrates that these barriers are solvable with modern machine learning and Python-based automation. By combining gradient boosting and LSTM neural networks for exposure forecasting, dynamic and ML-optimized hedge ratio methodologies, automated dual-standard effectiveness testing, and BCBS 368-integrated IRRBB reporting, the IDHF reduces the operational burden of derivative governance while materially improving hedging outcomes — a 34% reduction in residual FX variance, an 85% reduction in effectiveness testing time, and IRRBB compliance restoration in the simulated banking book case.

For U.S. regional banks, which face increasing supervisory scrutiny of their IRRBB frameworks following the 2023 banking stress, and for U.S. corporates navigating continued FX volatility and interest rate uncertainty, the deployment of integrated, automated hedging infrastructure is no longer a competitive advantage. It is the standard of responsible treasury governance.

The author intends to develop this framework into a deployable open-source toolkit, calibrated to U.S. regulatory requirements under ASC 815 and Federal Reserve IRRBB supervisory guidance, for distribution through the treasury and risk management professional community.

About the Author

Imran Siddiqui, CFA, has over twenty years of executive treasury leadership across financial institutions in the GCC and MENA region. He has designed and governed derivative hedging programs covering interest rate swaps, cross-currency swaps, FX forwards, options, par forwards, and structured collars. He has negotiated ISDA Master Agreements and Credit Support Annexures directly with international banking counterparties, overseen hedge accounting under both IFRS 9 and IAS 39 frameworks, and managed investment portfolios exceeding USD 3 billion in notional value. His computer science background enables direct Python development of the analytical systems described in this paper. He holds a Bachelor of Computer Science (Hamdard University), an MBA (SZABIST), and the CFA charter. He can be contacted through the CFA Institute membership directory.

Disclosure

The views expressed are those of the author in a personal professional capacity. Python implementations are illustrative and do not constitute production-ready or audited software. Empirical results are derived from synthetic datasets.

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